

Problem 1

Solution:

(a)

$$\binom{5}{3} = \frac{5!}{3!(5-3)!} = \frac{5!}{3!2!} = \frac{5(4)3!}{3!2!} = \frac{5(4)}{2(1)} = \frac{20}{2} = 10$$

(b)

$$\binom{7}{0} = \frac{7!}{0!(7-0)!} = \frac{7!}{0!7!} = \frac{7!}{7!} = 1$$

(c)

$$\binom{7}{1} = \frac{7!}{1!(7-1)!} = \frac{7!}{1!6!} = \frac{7!}{6!} = \frac{(7)6!}{6!} = 7$$

(d)

$$\binom{7}{2} = \frac{7!}{2!5!} = \frac{7(6)5!}{2!5!} = \frac{42}{2} = 21$$

(e)

$$\binom{17}{14} = \frac{17!}{14!3!} = \frac{17(16)(15)14!}{14!3!} = \frac{17(16)(15)}{6} = \frac{17(8)(5)}{1} = 680$$

(f)

$$\binom{0}{0} = \frac{0!}{0!0!} = 1$$

□

Problem 2

Solution:

(a)

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \frac{n!}{(n-k)!k!} = \binom{n}{n-k}$$

(b) We have $k = 10$ and $n - k = 7$. Hence, $n = 17$.

(c) We have $n = 14$ and $n - k = k - 4$. Thus $n = 2k - 4$ so $k = 9$.

(d) We have $n = 12$ and $n - k = k - 3$. Thus $n = 2k - 3$ so $k = \frac{15}{2}$. In other words, there is no integer k that works.

□

Problem 3

Solution:

$$\begin{aligned}
 \binom{n}{k-1} + \binom{n}{k} &= \frac{n!}{(k-1)!(n-(k-1))!} + \frac{n!}{k!(n-k)!} \\
 &= \frac{n!}{(k-1)!(n-k+1)!} + \frac{n!}{k!(n-k)!} \\
 &= \frac{n!}{(k-1)!(n-k+1)(n-k)!} + \frac{n!}{k(k-1)!(n-k)!} \\
 &= \frac{n!(k) + n!(n-k+1)}{k(k-1)!(n-k+1)(n-k)!} \\
 &= \frac{n![k+n-k+1]}{k(k-1)!(n-k+1)(n-k)!} \\
 &= \frac{n!(n+1)}{k(k-1)!(n-k+1)(n-k)!} \\
 &= \frac{(n+1)!}{k!((n-k+1)(n-k))!} \\
 &= \frac{(n+1)!}{k!((n+1)-k)!} \\
 &= \binom{n+1}{k}
 \end{aligned}$$

□

Problem 4

Solution:

For the base case ($n = 1$), observe that

$$(a+b)^1 = a+b = b+a = \binom{1}{0}a^0b^1 + \binom{1}{1}a^1b^0 = \sum_{k=0}^1 \binom{1}{k}a^kb^{1-k}$$

For the inductive hypothesis assume that for arbitrary fixed $q \geq 1$

$$(a+b)^q = \sum_{k=0}^q \binom{q}{k}a^kb^{q-k}$$

For the inductive step, we need to show

$$(a + b)^{q+1} = \sum_{k=0}^{q+1} \binom{q+1}{k} a^k b^{(q+1)-k}$$

Observe that

$$\begin{aligned} (a + b)^{q+1} &= (a + b)(a + b)^q \\ &= (a + b) \left(\sum_{k=0}^q \binom{q}{k} a^k b^{q-k} \right) \quad (\text{by inductive hypothesis}) \\ &= a \left[\sum_{k=0}^q \binom{q}{k} a^k b^{q-k} \right] + b \left[\sum_{k=0}^q \binom{q}{k} a^k b^{q-k} \right] \\ &= \left[\sum_{k=0}^q \binom{q}{k} a^{k+1} b^{q-k} \right] + \left[\sum_{k=0}^q \binom{q}{k} a^k b^{q-k+1} \right] \\ &= \left[\sum_{k=1}^{q+1} \binom{q}{k-1} a^k b^{q-(k-1)} \right] + \left[\sum_{k=0}^q \binom{q}{k} a^k b^{q-k+1} \right] \\ &= \left[\sum_{k=1}^{q+1} \binom{q}{k-1} a^k b^{q-k+1} \right] + \binom{q+1}{0} a^0 b^{q+1} + \left[\sum_{k=1}^q \binom{q}{k} a^k b^{q-k+1} \right] \\ &= \binom{q}{q} a^{q+1} b^0 + \left[\sum_{k=1}^q \binom{q}{k-1} a^k b^{q-k+1} \right] + \binom{q+1}{0} a^0 b^{q+1} + \left[\sum_{k=1}^q \binom{q}{k} a^k b^{q-k+1} \right] \\ &= \binom{q+1}{q+1} a^{q+1} b^0 + \binom{q+1}{0} a^0 b^{q+1} + \left[\sum_{k=1}^q \left(\binom{q}{k-1} + \binom{q}{k} \right) a^k b^{q-k+1} \right] \\ &= \binom{q+1}{q+1} a^{q+1} b^0 + \binom{q+1}{0} a^0 b^{q+1} + \left[\sum_{k=1}^q \binom{q+1}{k} a^k b^{q-k+1} \right] \quad (\text{by problem 3}) \\ &= \sum_{k=0}^{q+1} \binom{q+1}{k} a^k b^{(q+1)-k} \end{aligned}$$

Now that we have shown the primary theorem, observe that

$$2^n = (1 + 1)^n = \sum_{k=0}^n \binom{n}{k} 1^k 1^{n-k} = \sum_{k=0}^n \binom{n}{k}$$

Similarly,

$$0 = 0^n = ((-1) + 1)^n = \sum_{k=0}^n \binom{n}{k} (-1)^k 1^{n-k} = \sum_{k=0}^n \binom{n}{k} (-1)^k$$

□

Problem 5

Solution:

$$\prod_{k=1}^1 a_k = a_1$$

$$\prod_{k=1}^n a_k = a_n \left(\prod_{k=1}^{n-1} a_k \right) \text{ if } n > 1$$

□

Problem 6

Solution:

For the base case ($n = 1$), observe that

$$\prod_{k=1}^1 a_k b_k = a_1 b_1 = \left(\prod_{k=1}^1 a_k \right) b_1 = \left(\prod_{k=1}^1 a_k \right) \left(\prod_{k=1}^1 b_k \right)$$

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$

$$\prod_{k=1}^q a_k b_k = \left(\prod_{k=1}^q a_k \right) \left(\prod_{k=1}^q b_k \right)$$

Now, for the inductive step

$$\begin{aligned} \prod_{k=1}^{q+1} a_k b_k &= a_{q+1} b_{q+1} \left(\prod_{k=1}^q a_k b_k \right) && \text{(by defn. in problem 5)} \\ &= a_{q+1} b_{q+1} \left(\prod_{k=1}^q a_k \right) \left(\prod_{k=1}^q b_k \right) && \text{(by inductive hypothesis)} \\ &= \left[a_{q+1} \left(\prod_{k=1}^q a_k \right) \right] \left[b_{q+1} \left(\prod_{k=1}^q b_k \right) \right] \\ &= \left(\prod_{k=1}^{q+1} a_k \right) \left(\prod_{k=1}^{q+1} b_k \right) && \text{(by defn. in problem 5)} \end{aligned}$$

For the corollary, observe that

$$\prod_{k=1}^n c a_k = \left(\prod_{k=1}^n c \right) \left(\prod_{k=1}^n a_k \right) = c^n \left(\prod_{k=1}^n a_k \right)$$

□

Problem 7

Solution:

Assume all $a_k \neq 0$. For the base case ($n = 1$), observe that

$$\prod_{k=1}^1 \frac{a_k}{a_{k-1}} = \frac{a_1}{a_0}$$

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$

$$\prod_{k=1}^q \frac{a_k}{a_{k-1}} = \frac{a_q}{a_0}$$

Now, for the inductive step

$$\begin{aligned} \prod_{k=1}^{q+1} \frac{a_k}{a_{k-1}} &= \frac{a_{q+1}}{a_q} \left(\prod_{k=1}^q \frac{a_k}{a_{k-1}} \right) && \text{(by the defn. in problem 5)} \\ &= \frac{a_{q+1}}{a_q} \left(\frac{a_q}{a_0} \right) && \text{(by the inductive hypothesis)} \\ &= \frac{a_{q+1}}{a_0} \end{aligned}$$

□

Problem 8

Solution:

For the base case ($n = 1$), observe that

$$\prod_{k=1}^1 (1 + x^{2^{k-1}}) = 1 + x^{2^{1-1}} = 1 + x^{2^0} = 1 + x = \frac{(1+x)(1-x)}{1-x} = \frac{1-x^2}{1-x} = \frac{1-x^{2^1}}{1-x}$$

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$

$$\prod_{k=1}^q (1 + x^{2^{k-1}}) = \frac{1 - x^{2^q}}{1 - x}$$

Now, for the inductive step

$$\begin{aligned}
 \prod_{k=1}^{q+1} (1 + x^{2^{k-1}}) &= (1 + x^{2^q}) \left(\prod_{k=1}^q (1 + x^{2^{k-1}}) \right) && \text{(by the defn. in problem 5)} \\
 &= (1 + x^{2^q}) \left(\frac{1 - x^{2^q}}{1 - x} \right) \\
 &= \frac{1 - (x^{2^q})(x^{2^q})}{1 - x} \\
 &= \frac{1 - x^{2^q+2^q}}{1 - x} \\
 &= \frac{1 - x^{2(2^q)}}{1 - x} \\
 &= \frac{1 - x^{2^{q+1}}}{1 - x}
 \end{aligned}$$

When $x = 1$, we have

$$\prod_{k=1}^n (1 + x^{2^{k-1}}) = \prod_{k=1}^n (1 + 1^{2^{k-1}}) = \prod_{k=1}^n (1 + 1) = \prod_{k=1}^n 2 = 2^n$$

□

Problem 9

Solution:

The inequality only works in special cases, e.g. $0 \leq a_k < b_k$ or if $a_k < b_k \leq 0$ (and n is odd).

For example, the inequality holds in

$$(-6)(-5)(-4) = -120 < -60 = (-5)(-4)(-3)$$

but fails in

$$(-6)(-5) = 30 \not< 20 = (-5)(-4)$$

□

Problem 10

Solution:

For the base case ($n = 2$), observe that $x > 1 \implies x^2 = x(x) > 1(x) = x$. And if $0 < x < 1$ then $x^2 = x(x) < 1(x) = x$.

For the inductive hypothesis, assume that for arbitrary fixed $k \geq 2$, $x > 1 \implies x^k > x$ and $0 < x < 1 \implies x^k < x$

Now, for the inductive step, $x > 1 \implies x^{k+1} = x(x^k) > 1(x) = x$ since the inductive hypothesis gives us $x^k > x$. Now, if $0 < x < 1$, by the corresponding inductive hypothesis, we have $x^{k+1} = x(x^k) < 1(x) = x$.

□

Problem 11

Solution:

$n = 1$ fails because $2^1 = 2 \not< 1 = 1!$

$n = 2$ fails because $2^2 = 4 \not< 2 = 2!$

$n = 3$ fails because $2^3 = 8 \not< 6 = 3!$

We will show that $2^n < n!$ for all positive integers $n \geq 4$ by induction.

For the base case ($n = 4$), observe that $2^4 = 16 < 24 = 4(3)(2)1 = 4!$

For the inductive hypothesis, assume that for arbitrary fixed $k \geq 4$, $2^k < k!$

Now, for the inductive step, observe that $2^{k+1} = 2(2^k) < 5k! \leq (k+1)k! = (k+1)!$ ($2^k < k!$ comes from the inductive hypothesis and since $4 \leq k$ we have $5 \leq k+1$).

□

Problem 12

Solution:

(a) By the binomial theorem, we have

$$\begin{aligned}
 \left(1 + \frac{1}{n}\right)^n &= \left(\frac{1}{n} + 1\right)^n \\
 &= \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{n}\right)^k (1)^{n-k} \\
 &= \sum_{k=0}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \\
 &= \binom{n}{0} \left(\frac{1}{n}\right)^0 + \sum_{k=1}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \\
 &= 1 + \sum_{k=1}^n \binom{n}{k} \left(\frac{1}{n}\right)^k \\
 &= 1 + \sum_{k=1}^n \frac{n!}{k!(n-k)!} \left(\frac{1}{n}\right)^k \\
 &= 1 + \sum_{k=1}^n \frac{1}{k!} \left[n(n-1) \cdots (n-k+1) \left(\frac{1}{n}\right)^k \right] \\
 &= 1 + \sum_{k=1}^n \frac{1}{k!} \left[\left(\frac{n}{n}\right) \left(\frac{n-1}{n}\right) \cdots \left(\frac{n-(k-1)}{n}\right) \right] \\
 &= 1 + \sum_{k=1}^n \frac{1}{k!} \left[\prod_{r=0}^{k-1} \left(\frac{n-r}{n}\right) \right] \\
 &= 1 + \sum_{k=1}^n \frac{1}{k!} \left[\prod_{r=0}^{k-1} \left(1 - \frac{r}{n}\right) \right]
 \end{aligned}$$

(b) Lemma I: $2n^n < (n+1)^n$ for all positive $n \geq 2$.

Proof:

$$\begin{aligned}
(n+1)^n &= \sum_{k=0}^n \binom{n}{k} n^k (1)^{n-k} && \text{(by binomial theorem)} \\
&= \sum_{k=0}^n \binom{n}{k} n^k \\
&= \left[\sum_{k=0}^{n-2} \binom{n}{k} n^k \right] + \binom{n}{n-1} n^{n-1} + \binom{n}{n} n^n \\
&= \left[\sum_{k=0}^{n-2} \binom{n}{k} n^k \right] + n(n^{n-1}) + n^n \\
&= \left[\sum_{k=0}^{n-2} \binom{n}{k} n^k \right] + 2n^n \\
&\geq 1 + 2n^n && \text{(since } n \geq 2\text{)} \\
&> 2n^n
\end{aligned}$$

By lemma I we have,

$$2 < \frac{(n+1)^n}{n^n} = \left(\frac{n+1}{n} \right)^n = \left(1 + \frac{1}{n} \right)^n$$

Lemma II: For all $k > 1$

$$\prod_{r=0}^{k-1} \left(1 - \frac{r}{n} \right) < 1$$

Proof: Observe that

$$\begin{aligned}
\prod_{r=0}^{k-1} \left(1 - \frac{r}{n} \right) &= \prod_{r=0}^{k-1} \left(\frac{n}{n} - \frac{r}{n} \right) \\
&= \prod_{r=0}^{k-1} \left(\frac{n-r}{n} \right) \\
&= \left(\frac{n-0}{n} \right) \left(\frac{n-1}{n} \right) \cdots \left(\frac{n-(k-1)}{n} \right) \\
&< 1(1) \cdots (1) && (< \text{ vs. } \leq \text{ since } n \geq 2, \text{ there are } \geq 3 \text{ terms}) \\
&= 1
\end{aligned}$$

By lemma II we have,

$$\left(1 + \frac{1}{n} \right)^n = 1 + \sum_{k=1}^n \frac{1}{k!} \left[\prod_{r=0}^{k-1} \left(1 - \frac{r}{n} \right) \right] \leq 1 + \sum_{k=1}^n \frac{1}{k!}$$

We use $\leq$ here vs. $<$ because if $n = 1$ then there are no product terms and the left and right sides are equal.

Lemma III: For all positive n

$$\sum_{k=1}^n \frac{1}{k!} < 2$$

Proof: $n = 1, 2, 3$ are illustrated below

$$\begin{aligned} \sum_{k=1}^1 \frac{1}{k!} &= 1 &< 2 \\ \sum_{k=1}^2 \frac{1}{k!} &= 1 + \frac{1}{2} = \frac{3}{2} &< 2 \\ \sum_{k=1}^3 \frac{1}{k!} &= 1 + \frac{1}{2} + \frac{1}{6} = \frac{5}{3} &< 2 \end{aligned}$$

Now, let $n \geq 4$. By problem 11, since we have $0 < 2^n < n!$ for $n \geq 4$, it's also true that

$$\frac{1}{n!} < \frac{1}{2^n}$$

for $n \geq 4$. Therefore, we have

$$\begin{aligned} \sum_{k=1}^n \frac{1}{k!} &= 1 + \frac{1}{2!} + \frac{1}{3!} + \left(\sum_{k=4}^n \frac{1}{k!} \right) \\ &< 1 + \frac{1}{2} + \frac{1}{6} + \left(\sum_{k=4}^n \frac{1}{2^k} \right) \\ &= \frac{5}{3} + \left(\sum_{k=4}^n \frac{1}{2^k} \right) \end{aligned}$$

Now, observe that since

$$\frac{1}{2^k} = \frac{2}{2^k} - \frac{1}{2^k} = \frac{1}{2^{k-1}} - \frac{1}{2^k}$$

we have

$$\sum_{k=4}^n \frac{1}{2^k} = \sum_{k=4}^n \left(\frac{1}{2^{k-1}} - \frac{1}{2^k} \right) = \left(\frac{1}{2^3} - \frac{1}{2^4} \right) + \left(\frac{1}{2^4} - \frac{1}{2^5} \right) + \cdots + \left(\frac{1}{2^{n-1}} - \frac{1}{2^n} \right) = \frac{1}{2^3} - \frac{1}{2^n}$$

and therefore

$$\sum_{k=1}^n \frac{1}{k!} < \frac{5}{3} + \left(\sum_{k=4}^n \frac{1}{2^k} \right) = \frac{5}{3} + \left(\frac{1}{2^3} - \frac{1}{2^n} \right) < \frac{5}{3} + \frac{1}{2^3} = \frac{43}{24} < \frac{48}{24} = 2$$

Now, by lemma III, we also have

$$1 + \sum_{k=1}^n \frac{1}{k!} < 1 + 2 = 3$$

Thus, we have successfully shown that

$$2 < \left(1 + \frac{1}{n} \right)^n < 1 + \sum_{k=1}^n \frac{1}{k!} < 3$$

□

Problem 13

Solution:

(a) Observe that

$$\begin{aligned} (b-a)(b^{p-1} + b^{p-2}a + \cdots + ba^{p-2} + a^{p-1}) &= (b-a) \left(\sum_{k=0}^{p-1} a^k b^{p-1-k} \right) \\ &= b \left(\sum_{k=0}^{p-1} a^k b^{p-1-k} \right) - a \left(\sum_{k=0}^{p-1} a^k b^{p-1-k} \right) \\ &= \left(\sum_{k=0}^{p-1} a^k b^{p-k} \right) - \left(\sum_{k=0}^{p-1} a^{k+1} b^{p-(k+1)} \right) \\ &= \left(\sum_{k=0}^{p-1} a^k b^{p-k} \right) - \left(\sum_{k=1}^p a^k b^{p-k} \right) \\ &= a^0 b^{p-0} + \left(\sum_{k=1}^{p-1} a^k b^{p-k} \right) - \left(\left(\sum_{k=1}^{p-1} a^k b^{p-k} \right) + a^p b^{p-p} \right) \\ &= b^p - a^p \end{aligned}$$

(b) We will start by showing the righthand side. By part (a), we have

$$\begin{aligned} (n+1)^{p+1} - n^{p+1} &= ((n+1) - n)((n+1)^p + (n+1)^{p-1}n + \cdots + (n+1)^{p-p}n^p) \\ &= \sum_{k=0}^p (n+1)^{p-k} n^k \end{aligned}$$

Now, observe that $n < n + 1 \implies n^k < (n + 1)^k \implies (n + 1)^{p-k}n^k < (n + 1)^{p-k}(n + 1)^k$ for any arbitrary fixed k, p since n is positive. As a result, we have the following

$$\begin{aligned}(n + 1)^{p+1} - n^{p+1} &= \sum_{k=0}^p (n + 1)^{p-k}n^k \\ &< \sum_{k=0}^p (n + 1)^{p-k}(n + 1)^k \\ &= \sum_{k=0}^p (n + 1)^p \\ &= (p + 1)(n + 1)^p\end{aligned}$$

Note that we can claim $<$ vs. $\leq$ because p is positive. Hence, there is at least one term like $(n + 1)^{p-k}n^k < (n + 1)^{p-k}(n + 1)^k$ available in the summation.

Now, this implies that

$$\frac{(n + 1)^{p+1} - n^{p+1}}{p + 1} < (n + 1)^p$$

The lefthand side is similar. Since n, p are positive we have $n^{p-k}n^k < (n + 1)^{p-k}n^k$ and

$$\begin{aligned}(n + 1)^{p+1} - n^{p+1} &= \sum_{k=0}^p (n + 1)^{p-k}n^k \\ &> \sum_{k=0}^p n^{p-k}n^k \\ &= \sum_{k=0}^p n^p \\ &= (p + 1)n^p\end{aligned}$$

which implies

$$n^p < \frac{(n + 1)^{p+1} - n^{p+1}}{p + 1}$$

When put together, we get

$$n^p < \frac{(n + 1)^{p+1} - n^{p+1}}{p + 1} < (n + 1)^p$$

as desired.

(c) We will induct on n . For the base case ($n = 1$), observe that

$$\sum_{k=1}^{1-1} k^p = 0 < \frac{1}{p+1} = \frac{1^{p+1}}{p+1} < 1 = \sum_{k=1}^n k^p$$

For the inductive hypothesis, assume that for an arbitrary fixed $q \geq 1$

$$\sum_{k=1}^{q-1} k^p < \frac{q^{p+1}}{p+1} < \sum_{k=1}^q k^p$$

For the inductive step, we need to show

$$\sum_{k=1}^q k^p < \frac{(q+1)^{p+1}}{p+1} < \sum_{k=1}^{q+1} k^p$$

Observe that

$$\begin{aligned} \sum_{k=1}^q k^p &= \left(\sum_{k=1}^{q-1} k^p \right) + q^p \\ &< \frac{q^{p+1}}{p+1} + q^p && \text{(by inductive hypothesis)} \\ &< \frac{q^{p+1}}{p+1} + \frac{(q+1)^{p+1} - q^{p+1}}{p+1} && \text{(by part (b))} \\ &= \frac{(q+1)^{p+1}}{p+1} \end{aligned}$$

So we have shown the left half of the inequality. For the right, observe that

$$\begin{aligned} \frac{(q+1)^{p+1}}{p+1} &= \frac{q^{p+1}}{p+1} + \frac{(q+1)^{p+1} - q^{p+1}}{p+1} \\ &< \left(\sum_{k=1}^q k^p \right) + \frac{(q+1)^{p+1} - q^{p+1}}{p+1} && \text{(by induction hypothesis)} \\ &< \left(\sum_{k=1}^q k^p \right) + (q+1)^p && \text{(by part (b))} \\ &= \sum_{k=1}^{q+1} k^p \end{aligned}$$

so we have successfully shown

$$\sum_{k=1}^q k^p < \frac{(q+1)^{p+1}}{p+1} < \sum_{k=1}^{q+1} k^p$$

□

Problem 14

Solution:

For the base case ($n = 1$), observe that $1 + a_1 \geq 1 + a_1$

For the inductive hypothesis, assume that for fixed arbitrary $k \geq 1$,

$$(1 + a_1)(1 + a_2) \cdots (1 + a_k) \geq 1 + a_1 + a_2 + \cdots + a_k$$

Now, for the inductive step, we have

$$\begin{aligned} (1 + a_1) \cdots (1 + a_{k+1}) &\geq (1 + a_1 + a_2 + \cdots + a_k)(1 + a_{k+1}) && \text{(by inductive hypothesis)} \\ &= \left(1 + \sum_{j=1}^k a_j\right) + a_{k+1} \left(1 + \sum_{j=1}^k a_j\right) \\ &\geq \left(1 + \sum_{j=1}^k a_j\right) + a_{k+1} \\ &= 1 + a_1 + a_2 + \cdots + a_{k+1} \end{aligned}$$

For the second task, observe that for $k > 0$ (i.e. $k + 1 > 1$), we have

$$\begin{aligned} (1 + x)^{k+1} &= (1 + x)(1 + x)^k \\ &\geq (1 + x)(1 + kx) && \text{(by theorem 1.25)} \\ &= 1 + kx + x + kx^2 \\ &= [1 + (k + 1)x] + kx^2 \end{aligned}$$

which implies that

$$(1 + x)^{k+1} - [1 + (k + 1)x] \geq kx^2$$

Hence, if $(1 + x)^{k+1} = 1 + (k + 1)x$, we would have $0 \geq kx^2$. But since $k > 0$ and x^2 is never negative, kx^2 can never be negative. Therefore, we must have $kx^2 = 0 \implies x^2 = 0 \implies x = 0$.

□

Problem 15

Solution:

We will use induction.

For the base case ($n = 2$), observe that $1 \leq \frac{2}{2} < 2$. Hence $\left[\frac{2}{2}\right] = 1$. So we have

$$\frac{2!}{2^2} = \frac{2}{4} = \frac{1}{2} \leq \frac{1}{2} = \left(\frac{1}{2}\right)^1 = \left(\frac{1}{2}\right)^{\left[\frac{2}{2}\right]}$$

For the inductive hypothesis, assume that for arbitrary fixed $k \geq 2$,

$$\frac{k!}{k^k} \leq \left(\frac{1}{2}\right)^{\left[\frac{k}{2}\right]}$$

Now, we will show a few lemmas which will become useful in the inductive step

Lemma I: for all integers $k \geq 2$, $(k+1)^k \geq 2k^k$.

Proof:

$$\begin{aligned} 2 &< \left(1 + \frac{1}{k}\right)^k && \text{(for all integer } k \geq 2 \text{ by problem 12(b))} \\ &= \left(\frac{k+1}{k}\right)^k \\ &= \frac{(k+1)^k}{k^k} \end{aligned}$$

so we get $(k+1)^k > 2k^k$

Lemma II: for all integers

$$\left(\frac{1}{2}\right)^{\left[\frac{k}{2}\right]+1} \leq \left(\frac{1}{2}\right)^{\left[\frac{(k+1)}{2}\right]}$$

Proof: Let p be the greatest integer in $\frac{k}{2}$. Then, we have two cases.

If k is even, then there exists an integer q such that $2q = k$. But this integer must be p because $p < q \implies p+1 \leq q = \frac{k}{2}$, which contradicts the definition of the greatest integer (i.e. that p is the unique integer such that $p \leq \frac{k}{2} < p+1$). Now, since $2p = k$ we have $p = \frac{k}{2} < \frac{k+1}{2}$ and $p+1 = \frac{k}{2} + 1 = \frac{k+2}{2} > \frac{k+1}{2}$. This can be rewritten as $p < \frac{k+1}{2} < p+1$, i.e. p is also the greatest integer in $\frac{k+1}{2}$. Therefore, we have

$$\left[\frac{k}{2}\right] = \left[\frac{k+1}{2}\right] \implies \left[\frac{k}{2}\right] + 1 > \left[\frac{k+1}{2}\right] \implies \left(\frac{1}{2}\right)^{\left[\frac{k}{2}\right]+1} < \left(\frac{1}{2}\right)^{\left[\frac{(k+1)}{2}\right]}$$

since $0 < \frac{1}{2} < 1$.

If k is odd, then by definition $k + 1$ is even. That is, there exists an integer q such that $2q = k + 1$. Now, observe that $q - 1 = \frac{k+1}{2} - 1 = \frac{k-1}{2}$. Hence, $q - 1 = \frac{k-1}{2} < \frac{k}{2} < \frac{k+1}{2} = q$, i.e. $q - 1$ is the greatest integer in $\frac{k+1}{2}$. But that means that $q - 1 = p!$ Also, since $q = \frac{k+1}{2}$, we have $\frac{k+1}{2} < q + 1 = p + 2$. Putting it all together, we see that $p + 1 < \frac{k+1}{2} < p + 2$, i.e. $p + 1$ is the greatest integer in $\frac{k+1}{2}$. Therefore,

$$\left\lfloor \frac{k}{2} \right\rfloor + 1 = \left\lfloor \frac{k+1}{2} \right\rfloor \implies \left(\frac{1}{2} \right)^{\left\lfloor \frac{k}{2} \right\rfloor + 1} = \left(\frac{1}{2} \right)^{\left\lfloor \frac{k+1}{2} \right\rfloor}$$

Hence, in both cases

$$\left(\frac{1}{2} \right)^{\left\lfloor \frac{k}{2} \right\rfloor + 1} \leq \left(\frac{1}{2} \right)^{\left\lfloor \frac{k+1}{2} \right\rfloor}$$

Now we can proceed with the inductive step. Observe that

$$\begin{aligned} \frac{(k+1)!}{(k+1)^{k+1}} &= \frac{(k+1)k!}{(k+1)(k+1)^k} \\ &= \frac{k!}{(k+1)^k} \\ &\leq \frac{k!}{2k^k} \quad (\text{lemma I gives us } (k+1)^k \geq 2k^k \implies \frac{1}{(k+1)^k} \leq \frac{1}{2k^k}) \\ &\leq \left(\frac{1}{2} \right) \left(\frac{1}{2} \right)^{\left\lfloor \frac{k}{2} \right\rfloor} \quad (\text{by inductive hypothesis}) \\ &= \left(\frac{1}{2} \right)^{\left\lfloor \frac{k}{2} \right\rfloor + 1} \\ &\leq \left(\frac{1}{2} \right)^{\left\lfloor \frac{k+1}{2} \right\rfloor} \quad (\text{by lemma II}) \end{aligned}$$

This closes the induction. □

Problem 16

Solution:

For the first base case ($n = 1$), observe that

$$a_1 = 1 = \frac{2}{2} < \frac{3}{2} = \frac{1+2}{2} = \frac{1+\sqrt{4}}{2} < \frac{1+\sqrt{5}}{2} = \left(\frac{1+\sqrt{5}}{2} \right)^1$$

For the second base case ($n = 2$), observe that

$$\begin{aligned}
 \left(\frac{1+\sqrt{5}}{2}\right)^2 &= \left(\frac{1+\sqrt{5}}{2}\right) \left(\frac{1+\sqrt{5}}{2}\right) \\
 &= \left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right) \left(\frac{1}{2} + \frac{\sqrt{5}}{2}\right) \\
 &= \left(\frac{1}{2}\right)^2 + 2\left(\frac{\sqrt{5}}{4}\right) + \left(\frac{\sqrt{5}}{2}\right)^2 \\
 &= \frac{1}{4} + \frac{\sqrt{5}}{2} + \frac{5}{4} \\
 &= \frac{3}{2} + \frac{\sqrt{5}}{2} \\
 &= \frac{3+\sqrt{5}}{2}
 \end{aligned}$$

$$a_2 = 2 = \frac{4}{2} < \frac{5}{2} = \frac{3+2}{2} = \frac{3+\sqrt{4}}{2} < \frac{3+\sqrt{5}}{2} = \left(\frac{1+\sqrt{5}}{2}\right)^2$$

Proof of the third base case ($n = 3$) is omitted since it's identical to the other two.

For the inductive hypothesis, we will use strong induction.

Assume that for any arbitrary fixed $q \geq 3$ such that $1 \leq q < k$

$$a_q < \left(\frac{1+\sqrt{5}}{2}\right)^q$$

Now, for the inductive step,

$$\begin{aligned}
a_k &= a_{k-1} + a_{k-2} \\
&< \left(\frac{1+\sqrt{5}}{2}\right)^{k-1} + \left(\frac{1+\sqrt{5}}{2}\right)^{k-2} && \text{(by inductive hypothesis)} \\
&= \left(\frac{1+\sqrt{5}}{2}\right) \left(\frac{1+\sqrt{5}}{2}\right)^{k-2} + \left(\frac{1+\sqrt{5}}{2}\right)^{k-2} \\
&= \left(\frac{1+\sqrt{5}}{2}\right)^{k-2} \left[\left(\frac{1+\sqrt{5}}{2}\right) + 1 \right] \\
&= \left(\frac{1+\sqrt{5}}{2}\right)^{k-2} \left(\frac{1+\sqrt{5}+2}{2}\right) \\
&= \left(\frac{1+\sqrt{5}}{2}\right)^{k-2} \left(\frac{3+\sqrt{5}}{2}\right) \\
&= \left(\frac{1+\sqrt{5}}{2}\right)^{k-2} \left(\frac{1+\sqrt{5}}{2}\right)^2 && \text{(by the second base case)} \\
&= \left(\frac{1+\sqrt{5}}{2}\right)^k
\end{aligned}$$

□

Problem 17

Solution:

By the Cauchy-Schwartz inequality with $a_k = x_k^p$ and $b_k = 1$, we have

$$\left(\sum_{k=1}^n x_k^p\right)^2 = \left(\sum_{k=1}^n x_k^p(1)\right)^2 \leq \left(\sum_{k=1}^n (x_k^p)^2\right) \left(\sum_{k=1}^n (1)^2\right) = \left(\sum_{k=1}^n x_k^{2p}\right) n$$

where

$$0 < \left(\sum_{k=1}^n x_k^p\right)^2$$

Also, since not all x_k are equal there cannot be a real number q such that $x_k^p q + b_k = x_k^p q + 1 = 0$ for all k . Otherwise, we would have $x_k^p = \frac{-1}{q}$ for every k , or, in other words, all x_k^p equal to one another (contradiction). Hence, we can tighten $\leq$ to $<$.

Now,

$$\frac{1}{n} \left(\sum_{k=1}^n x_k^p \right)^2 < \left(\sum_{k=1}^n x_k^{2p} \right)$$

and thereby

$$\frac{1}{n^2} \left(\sum_{k=1}^n x_k^p \right)^2 = \left(\frac{\sum_{k=1}^n x_k^p}{n} \right)^2 < \left(\frac{\sum_{k=1}^n x_k^{2p}}{n} \right) = \frac{1}{n} \left(\sum_{k=1}^n x_k^{2p} \right)$$

Now, taking the $2p$ -th root of both sides gives

$$\left(\left(\frac{\sum_{k=1}^n x_k^p}{n} \right)^2 \right)^{\frac{1}{2p}} < \left(\frac{\sum_{k=1}^n x_k^{2p}}{n} \right)^{\frac{1}{2p}}$$

and finally

$$\left(\frac{\sum_{k=1}^n x_k^p}{n} \right)^{\frac{1}{p}} < \left(\frac{\sum_{k=1}^n x_k^{2p}}{n} \right)^{\frac{1}{2p}} \implies M_p < M_{2p}$$

□

Problem 18

Solution:

Since $a > 0$, $b > 0$, $c > 0$ and $p = 2 > 0$, we can invoke problem 17 to justify

$$\left(\frac{a^2 + b^2 + c^2}{3} \right)^{\frac{1}{2}} \leq \left(\frac{a^4 + b^4 + c^4}{3} \right)^{\frac{1}{2(2)}}$$

We're using $\leq$ vs. $<$ since a, b, c could all be equal.

Then,

$$\left(\frac{a^2 + b^2 + c^2}{3} \right)^{\frac{1}{2}} = \left(\frac{8}{3} \right)^{\frac{1}{2}} \leq \left(\frac{a^4 + b^4 + c^4}{3} \right)^{\frac{1}{2(2)}}$$

By taking the fourth power of both sides, we get

$$\left(\left(\frac{8}{3} \right)^{\frac{1}{2}} \right)^4 = \left(\frac{8}{3} \right)^2 = \frac{64}{9} \leq \left(\frac{a^4 + b^4 + c^4}{3} \right) = \left(\left(\frac{a^4 + b^4 + c^4}{3} \right)^{\frac{1}{4}} \right)^4$$

This implies that

$$\left(\frac{64}{9}\right)(3) \leq a^4 + b^4 + c^4 \implies a^4 + b^4 + c^4 \geq \frac{64}{3}$$

□

Problem 19

Solution:

We have two cases. If all $a_k = 1$ then $a_1 + a_2 + \cdots + a_n = 1 + \cdots + 1 = 1(n) = n \geq n$.

Now, suppose there exists at least one $a_k \neq 1$. We will use induction to show that if we have $a_1 \cdot a_2 \cdots a_n = 1$ we must still have $a_1 + a_2 + \cdots + a_n \geq n$ for all $n \geq 1$.

For the base case ($n = 1$) suppose that $a_1 \neq 1$. Then, the product of all the a_k is not 1 so the proposition we're trying to show (namely $a_1 = 1 \implies a_1 \geq 1$) is vacuously true.

For the inductive hypothesis, assume that for arbitrary fixed $q \geq 1$, if we have $a_1 \cdot a_2 \cdots a_q = 1$ then we must also have $a_1 + a_2 + \cdots + a_q \geq q$.

Now, for the inductive step, we have a_k such that $a_1 \cdot a_2 \cdots a_q \cdot a_{q+1} = 1$ and at least one $a_k \neq 1$. Observe that there must then be one $a_k > 1$ and another < 1 ; if there is no $a_k < 1$ then the product of all the a_k will be > 1 (contradiction) and otherwise, if there is no $a_k > 1$ the product of all the a_k will be < 1 (contradiction again). Without loss of generality let $a_i < 1$ and $a_j > 1$ ($i \neq j$) and let $b = a_i \cdot a_j$. Then, we can rewrite the product as $b \left(\prod_{1 \leq k \leq (q+1), k \neq i, k \neq j} a_k \right) = 1$.

Now, by the inductive hypothesis (since we have q total positive reals whose product = 1), we can claim that

$$b + \left(\sum_{1 \leq k \leq (q+1), k \neq i, k \neq j} a_k \right) \geq q$$

Now, observe that $a_i < 1 \wedge a_j > 1 \implies (a_i - 1)(a_j - 1) < 0$. Therefore $(a_i - 1)(a_j - 1) = a_i a_j - a_i - a_j + 1 = b - a_i - a_j + 1 < 0$. This in turn implies, however, that $b < a_i + a_j - 1$. As a result, we have

$$\begin{aligned}
 q &\leq b + \left(\sum_{1 \leq k \leq (q+1), k \neq i, k \neq j} a_k \right) \\
 &< a_i + a_j - 1 + \left(\sum_{1 \leq k \leq (q+1), k \neq i, k \neq j} a_k \right) \\
 &= \left(a_i + a_j + \left(\sum_{1 \leq k \leq (q+1), k \neq i, k \neq j} a_k \right) \right) - 1 \\
 &= \left(\sum_{k=1}^{q+1} a_k \right) - 1
 \end{aligned}$$

which implies that

$$\sum_{k=1}^{q+1} a_k \geq q + 1$$

This closes the induction and concludes the proof of the second case. Thus, in both cases, we showed that $a_1 \cdot a_2 \cdots a_n = 1$ implies $a_1 + a_2 + \cdots + a_n \geq n$ for positive reals a_k .

For the second part, we must further show that under these conditions $a_1 + a_2 + \cdots + a_n = n \iff a_1 = a_2 = \cdots = a_n = 1$. We'll show the reverse case first. Let $a_1 = a_2 = \cdots = a_n = 1$. Then, $a_1 + a_2 + \cdots + a_n = 1 + 1 + \cdots + 1 = 1(n) = n$.

Now, the forward case. Suppose by way of contradiction that $a_1 + a_2 + \cdots + a_n = n$ but not all $a_k = 1$. Then, similarly to the proof above, we know, without loss of generality, that there exist $a_i > 1$ and $a_j < 1$ such that $\prod_{t=1}^n a_t = 1$. Again, let $b = a_i a_j$ and observe that $(a_i - 1)(a_j - 1) = b - a_i - a_j + 1 < 0 \implies b < a_i + a_j - 1$. Now, by the prior proof, we have

$$b \left(\prod_{k \neq i, j} a_k \right) = 1 \implies b + \left(\sum_{k \neq i, j} a_k \right) \geq (n - 1)$$

However, from our inequality on b we have

$$(n - 1) \leq b + \left(\sum_{k \neq i, j} a_k \right) < a_i + a_j - 1 + \left(\sum_{k \neq i, j} a_k \right) = -1 + \left(\sum_{k=1}^n a_k \right) = n - 1 \implies n - 1 < n - 1$$

which is a clear contradiction. Hence, all the a_k must be $= 1$.

□

Problem 20

Solution:

(a) Observe that

$$\frac{x_1}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} \cdot \frac{x_2}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} \cdots \frac{x_n}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} = \frac{x_1 \cdot x_2 \cdots x_n}{((x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}})^n} = \frac{x_1 \cdot x_2 \cdots x_n}{x_1 \cdot x_2 \cdots x_n} = 1$$

Therefore, by problem 19

$$\frac{x_1}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} + \frac{x_2}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} + \cdots + \frac{x_n}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} \geq n$$

which implies

$$x_1 + x_2 + \cdots + x_n \geq n(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} \implies (x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} \leq \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Now, for the second part, we must show that

$$(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} = \frac{x_1 + x_2 + \cdots + x_n}{n} \iff x_1 = x_2 = \cdots = x_n$$

We'll begin with the forward case. Observe that

$$n = \frac{x_1 + x_2 + \cdots + x_n}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} = \frac{x_1}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} + \frac{x_2}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} + \cdots + \frac{x_n}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}}$$

so by problem 19 again, we know that

$$\frac{x_1}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} = \frac{x_2}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} = \cdots = \frac{x_n}{(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}} = 1$$

Therefore, it must be the case that $x_1 = x_2 = \cdots = x_n$.

For the reverse case, observe that if $q = x_1 = x_2 = \cdots = x_n$, then we have

$$\frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{q + q + \cdots + q}{n} = \frac{q(n)}{n} = q = (q^n)^{\frac{1}{n}} = (q \cdot q \cdots q)^{\frac{1}{n}} = (x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}$$

(b) For the first part, we must show that $G < M_p$, or, in another form,

$$(x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} < \left(\frac{x_1^p + x_2^p + \cdots + x_n^p}{n} \right)^{\frac{1}{p}}$$

Consider the reals x_k^p . They're positive since x_k are all positive. Also, since not all x_k are equal, not all x_k^p are equal. Thus, by part (a) we have

$$\begin{aligned} 0 < (x_1^p \cdot x_2^p \cdots x_n^p)^{\frac{1}{n}} < \frac{x_1^p + x_2^p + \cdots + x_n^p}{n} &\implies 0 < ((x_1 \cdot x_2 \cdots x_n)^p)^{\frac{1}{n}} < \frac{x_1^p + x_2^p + \cdots + x_n^p}{n} \\ &\implies 0 < ((x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}})^p < \frac{x_1^p + x_2^p + \cdots + x_n^p}{n} \\ &\implies 0 < (x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} < \left(\frac{x_1^p + x_2^p + \cdots + x_n^p}{n} \right)^{\frac{1}{p}} \end{aligned}$$

Last step follows because it can be shown that, given positive reals a, b and a positive rational $\frac{m}{n}$, $0 < a < b \implies 0 < a^{\frac{m}{n}} < b^{\frac{m}{n}}$. Proof is omitted, but left as an exercise to the reader.

For the second part, we must show that $M_q < G$, or, in another form,

$$\left(\frac{x_1^q + x_2^q + \cdots + x_n^q}{n} \right)^{\frac{1}{q}} < (x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}}$$

Consider the reals x_k^q . They're still positive since $x_k^q = \frac{1}{x_k^{-q}}$ ($q < 0 \implies -q > 0$). Also, since not all x_k are equal, not all x_k^q are equal. Thus, by part (a) we have

$$\begin{aligned} 0 < (x_1^q \cdot x_2^q \cdots x_n^q)^{\frac{1}{n}} < \frac{x_1^q + x_2^q + \cdots + x_n^q}{n} &\implies 0 < ((x_1 \cdot x_2 \cdots x_n)^q)^{\frac{1}{n}} < \frac{x_1^q + x_2^q + \cdots + x_n^q}{n} \\ &\implies 0 < ((x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}})^q < \frac{x_1^q + x_2^q + \cdots + x_n^q}{n} \\ &\implies 0 < \left(\frac{x_1^q + x_2^q + \cdots + x_n^q}{n} \right)^{\frac{1}{q}} < (x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} \end{aligned}$$

Last step follows because it can be shown that, given positive reals a, b and a negative integer k , $0 < a^k < b \implies 0 < b^{\frac{1}{k}} < a$. The proof is omitted but follows from two lemmas: $0 < a < b \implies 0 < b^{-1} < a^{-1}$ (proved in an earlier section) and the previously mentioned lemma $0 < a < b \implies 0 < a^{\frac{m}{n}} < b^{\frac{m}{n}}$ where $\frac{m}{n}$ is a positive rational.

□

Problem 21

Solution:

Since a, b, c are positive reals, by problem 20(a) we have

$$0 < (abc)^{\frac{1}{3}} \leq \frac{a + b + c}{3}$$

which simplifies to

$$8^{\frac{1}{3}} \leq \frac{a+b+c}{3} \implies a+b+c \geq 6$$

Now, by problem 20(b) we have the following ($q = -1$)

$$\begin{aligned} \left(\frac{a^{-1} + b^{-1} + c^{-1}}{3} \right)^{-1} &\leq (abc)^{\frac{1}{3}} \implies \frac{3}{a^{-1} + b^{-1} + c^{-1}} \leq 8^{\frac{1}{3}} \\ &\implies \frac{3}{a^{-1} + b^{-1} + c^{-1}} \leq 2 \\ &\implies \frac{3}{2} \leq a^{-1} + b^{-1} + c^{-1} \\ &\implies \frac{12}{8} \leq a^{-1} + b^{-1} + c^{-1} \\ &\implies 12 \leq 8(a^{-1} + b^{-1} + c^{-1}) \\ &\implies 12 \leq abc(a^{-1} + b^{-1} + c^{-1}) \\ &\implies 12 \leq bc + ac + ab \end{aligned}$$

We use $\leq$ because if $a = b = c$, then

$$\begin{aligned} \left(\frac{a^{-1} + b^{-1} + c^{-1}}{3} \right)^{-1} &= \frac{3}{a^{-1} + b^{-1} + c^{-1}} \\ &= \frac{3}{3a^{-1}} \\ &= \frac{1}{a^{-1}} \\ &= a \\ &= (a^3)^{\frac{1}{3}} \\ &= (abc)^{\frac{1}{3}} \end{aligned}$$

□

Problem 22

Solution:

Observe that

$$\begin{aligned}
1 &= \frac{x_1 \cdot x_2 \cdots x_n}{x_1 \cdot x_2 \cdots x_n} \\
&= (x_1 \cdot x_2 \cdots x_n) \left(\frac{1}{x_1} \right) \left(\frac{1}{x_2} \right) \cdots \left(\frac{1}{x_n} \right) \\
&= (x_1 \cdot x_2 \cdots x_n) \left(\frac{1}{x_1} \cdot \frac{1}{x_2} \cdots \frac{1}{x_n} \right) \\
&= \left[(x_1 \cdot x_2 \cdots x_n) \left(\frac{1}{x_1} \cdot \frac{1}{x_2} \cdots \frac{1}{x_n} \right) \right]^{\frac{1}{n}} \\
&= (x_1 \cdot x_2 \cdots x_n)^{\frac{1}{n}} \left(\frac{1}{x_1} \cdot \frac{1}{x_2} \cdots \frac{1}{x_n} \right)^{\frac{1}{n}} \\
&\leq \left(\frac{x_1 + x_2 + \cdots + x_n}{n} \right) \left(\frac{1}{x_1} \cdot \frac{1}{x_2} \cdots \frac{1}{x_n} \right)^{\frac{1}{n}} && \text{(by AM-GM inequality)} \\
&\leq \left(\frac{x_1 + x_2 + \cdots + x_n}{n} \right) \left(\frac{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}{n} \right) && \text{(by AM-GM inequality)}
\end{aligned}$$

which implies that

$$\begin{aligned}
n^2 &\leq (x_1 + x_2 + \cdots + x_n) \left(\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} \right) \\
&= \left(\sum_{k=1}^n x_k \right) \left(\sum_{k=1}^n \frac{1}{x_k} \right) \\
&= \left(\sum_{k=1}^n x_k \right) \left(\sum_{k=1}^n y_k \right)
\end{aligned}$$

□

Problem 23

Solution:

From problem 22 we have

$$(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9 = 3^2$$

But since $a + b + c = 1$, this simplifies to

$$\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) = \frac{bc}{abc} + \frac{ac}{abc} + \frac{ab}{abc} = \frac{ab + ac + bc}{abc} \geq 9 \implies ab + ac + bc \geq 9abc$$

which in turn gives us

$$\begin{aligned} ab + ac + bc - abc &\geq 8abc \implies [1 - (a + b + c)] + [(ab + ac + bc) - abc] \geq 8abc \\ &\implies (1 - a)(1 - b)(1 - c) \geq 8abc \end{aligned}$$

□